Computer Vision Practical Report:

EDGE DETECTION USING SOBEL MATRIX ALONG XY AXIS

1. AIM

To detect both horizontal and vertical edges in an image by combining Sobel gradients along the X and Y axes using OpenCV, display the results, and save the output image.

2. PROCEDURE

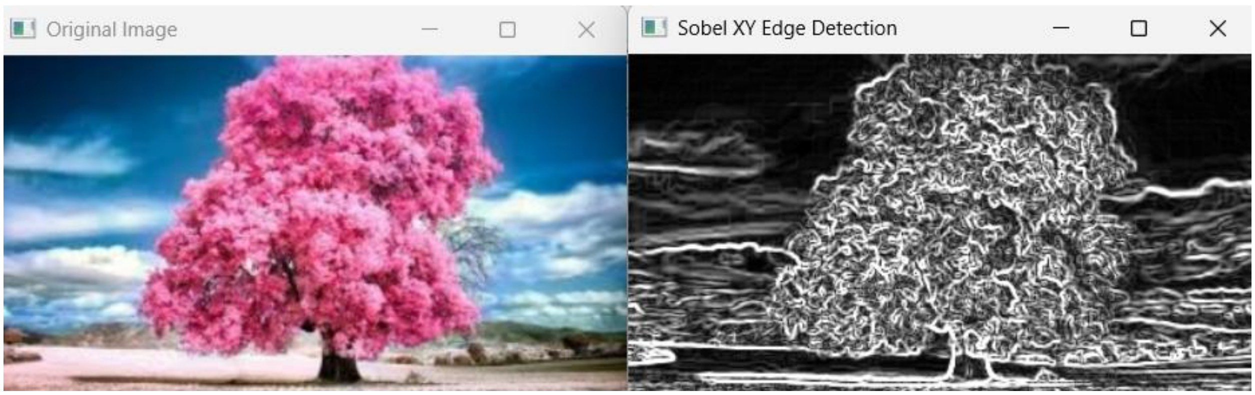
- Load the image and convert it to grayscale.
- Apply Gaussian blur to reduce noise.
- Compute horizontal gradients (dx=1, dy=0) and vertical gradients (dx=0, dy=1) using cv2.Sobel().
- Convert both gradient results to 8-bit format using cv2.convertScaleAbs().
- Combine both gradient images using cv2.addWeighted() to obtain the final edge magnitude.
- Display the original and combined edge images using Matplotlib, and save the result using cv2.imwrite().

3. PROGRAM CODE

```
import cv2

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E19-IMG.png")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_xy = cv2.magnitude(sobel_x, sobel_y)
sobel_xy = cv2.convertScaleAbs(sobel_xy)
```

```
cv2.imshow("Original Image", image)
cv2.imshow("Sobel XY Edge Detection", sobel_xy)

cv2.imwrite("sobel_xy.jpg", sobel_xy)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.